

# 1 Sup Inf & Set || Nested Intervals || Sequence & Bolzano Weierstrass

## 1.1 Supremum and Infimum & Set

**Completeness Axiom:** If  $\exists \text{ UP} \Rightarrow \text{LUB} (\sup)$  ; If  $\exists \text{ LB} \Rightarrow \text{GLB} (\inf)$ . **Archimedean Property:**  $\forall x \in \mathbb{R}^+ \Rightarrow \exists n \in \mathbb{N} \text{ s.t. } n > x$ .

**Countable Set:** A set  $E$  is countable if  $\exists$  bijection  $f : \mathbb{N} \rightarrow E$ . ps:  $[0, 1]$  is uncountable.

· **Property:** Let  $(E_n)_{n \in \mathbb{N}}$  be countable sets.  $\Rightarrow {}^1 \cup E_n$  is countable. (Can infinite)  ${}^2 \prod E_n$  is countable. (Finite)  ${}^3 [0, 1]$  uncountable.

${}^\oplus$  **Dense Set:** Set  $A$  is dense in  $\mathbb{R}$  iff  $\forall x \in \mathbb{R}, \forall \varepsilon > 0 \exists a \in A \text{ s.t. } |x - a| < \varepsilon$  e.g.  ${}^\oplus \mathbb{Q}$  is dense in  $\mathbb{R}$

${}^\oplus$  **Power Set  $\mathcal{P}$ :**  $\mathcal{P}(E)$  is the set of all subsets of  $E$ . ps:  $\mathcal{P}(\mathbb{R})$  is the set of all subsets of  $\mathbb{R}$ .

**Property of Sup , Inf:**  $\inf(-f_n) = -\sup(f_n)$  ||  $\limsup(-f_n) = -\liminf(f_n)$  || If  $g$  is fixed,  $\sup(f_n + g) = \sup(f_n) + g$  ps:  $\forall \inf, \limsup, \liminf$  也成立

## 1.2 Nested Intervals

**Def of  $\lambda(I)$ :** Let  $I$  be an interval.  $\lambda(I) := \sup I - \inf I$  ||  $\lambda((a, b)) = \lambda([a, b)) = \lambda((a, b]) = \lambda([a, b]) = b - a$ .

**Nested Intervals:** Sequence  $(I_n)_{n \in \mathbb{N}}$  is nested if  $I_n \supseteq I_{n+1} (\downarrow)$  for all  $n \in \mathbb{N}$ .

**Nested Interval Theorem:** If  $(I_n)_{n \in \mathbb{N}}$  is nested *closed bounded* interval sequence  $\Rightarrow E := \bigcap I_n \neq \emptyset$ .

· **property:**  ${}^1$  If  $\lim \lambda(I_n) = 0 \Rightarrow E = a$   ${}^2 \oplus$  If  $I_n = [a_n, b_n] \Rightarrow E = [\sup a_n, \inf b_n]$

## 1.3 Sequence, Cauchy & Bolzano Weierstrass Theorem

**Useful Seq Limit:**  $\lim \frac{1}{n^p} = 0, p > 1$  ;  $\lim a^n = 0, |a| < 1$  ;  $\lim n^{\frac{1}{n}} = 1$  ;  $\lim a^{\frac{1}{n}} = 1, a > 0$

**Operation in Seq Limit:** *Squeeze Theorem ; Monotone Convergence Theorem (MCT)* ;  $\lim ax_n = a \lim x_n$

·  $\lim(x_n + y_n) = \lim x_n + \lim y_n$  ;  $\lim x_n \cdot y_n = \lim x_n \cdot \lim y_n$  ;  $\lim \frac{x_n}{y_n} = \frac{\lim x_n}{\lim y_n}, y_n \neq 0$

**Comparison Theorem (For Seq):** If  $\lim x_n, y_n$  exist, and  $x_n \leq y_n$  for all  $n \geq N$  then  $\lim x_n \leq \lim y_n$ .

**Cauchy Sequence:** Sequence  $(a_n)_{n \in \mathbb{N}}$  is Cauchy iff  $\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } |a_n - a_m| < \varepsilon$  for all  $m, n \geq N$ .

**Cauchy Criterion:** Convergent  $\Leftrightarrow$  Cauchy proof:  $(\Rightarrow) |a_n - a_m| \leq |a_n - a| + |a - a_m|$   $(\Leftarrow) a_n$  converge  $\rightarrow$  bdd; Bolzano-Weierstrass  $\rightarrow a_{n_k} \rightarrow a$  converge;  $|a_n - a| \leq |a_n - a_{n_k}| + |a_{n_k} - a|$

Proof of Uniformly Cauchy  $\Leftrightarrow$  Uniformly Convergent.  $(\Rightarrow)$  unif cauchy  $\rightarrow$  pointwise cauchy  $\rightarrow$  pointwise convergent  $\rightarrow \forall x \in E, \lim f_n(x) = f(x); |f_n(x) - f_m(x)| < \varepsilon/2 \rightarrow \lim_m |f_n(x) - f(x)| \leq \varepsilon/2 < \varepsilon (\Leftarrow) |f_n(x) - f_m(x)| \leq |f_n(x) - f(x)| + |f(x) - f_m(x)|$

**Bolzano-Weierstrass Theorem:** Every *bounded sequence* has a *convergent subsequence*.

**Def of limit superior and inferior:**  $\limsup a_n := \lim(\sup_{k \geq n} a_k)$  ||  $\liminf a_n := \lim(\inf_{k \geq n} a_k)$

· **Property:** If  $(a_n)_{n \in \mathbb{N}}$  bounded.  $\Rightarrow {}^\oplus {}^1 \exists(a_{n_k}) \rightarrow \limsup a_n$  Hint:  $\exists a_{n_k}, \sup_{j \geq k} a_j - \frac{1}{k} \leq a_{n_k} \leq \sup_{j \geq k} a_j, \sup_{j \geq n_k} a_j \leq \sup_{j \geq k} a_j$   ${}^\oplus {}^2 \exists(a_{n_k}) \rightarrow \liminf a_n$   
 ${}^\oplus {}^3 \forall a_{n_k} \rightarrow a \liminf a_n \leq a \leq \limsup a_n$

**Subsequence Theorem:** In each case, the subsequence can be taken to be *monotonic*.

- $\exists a \in \mathbb{R} \text{ s.t. } \forall \varepsilon > 0$ , there are infinitely many  $n \in \mathbb{N} \text{ s.t. } |x_n - a| < \varepsilon$  ( $\exists$  聚点  $a$ )  $\Leftrightarrow \exists$  subsequence of  $(x_n) \text{ s.t. } \lim x_{n_k} = a$
- $(x_n)_{n \in \mathbb{N}}$  is not bounded above (below)  $\Leftrightarrow \exists$  subsequence of  $x_n \text{ s.t. } \lim x_{n_k} = \infty (-\infty)$
- If  $\lim x_n = L \Rightarrow \forall x_{n_k}, \lim x_{n_k} = L$

## 2 Series

**Cauchy Criterion for Series:**  $\sum_{k=1}^{\infty} a_k$  converges  $\Leftrightarrow \forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq m \geq N | \sum_{k=m+1}^n a_k | = |a_{m+1} + \dots + a_n| < \varepsilon$ .

**p-Test:**  $\sum \frac{1}{n^p}$  convergent iff  $p > 1$  **Root Test** **Ratio Test**  $_{|n|^{1/n}, |n+1/n|}$  **Geometric Series:**  $\sum p^n$  converges iff  $|p| < 1$

**Divergence Test:** If  $\lim a_n \neq 0 \Rightarrow \sum a_n$  diverges. (If  $\sum a_n$  convergent  $\Rightarrow \lim a_n = 0$ .)

**Comparison Test:** If  $0 < a_n < b_n, \sum b_n$  convergent  $\Rightarrow \sum a_n$  also ;  $\sum a_n$  divergent  $\Rightarrow \sum b_n$  also.

· **Limit Comparison Test:** If  $a_n, b_n \geq 0, \lim \frac{a_n}{b_n} = L > 0 \Rightarrow$  If  $L \in (0, \infty)$ :  $\sum a_n$  convergent iff  $\sum b_n$  also.

If  $L = 0$  :  $\sum b_n$  convergent  $\Rightarrow \sum a_n$  also.  $L = \infty$  :  $\sum b_n$  divergent  $\Rightarrow \sum a_n$  also.

**Integral Test:** Let  $f : [1, \infty) \rightarrow \mathbb{R}$  is 非负递减,  $a_n = f(n)$ . Then  $\sum a_n$  converges iff  $\int_1^{\infty} f(x) dx < \infty$ .

**Absolutely Convergence:**  $\sum a_n$  convergent absolutely iff  $\sum |a_n|$  convergent. **If convergent abs  $\Rightarrow$  convergent.**

**Alternating Series Test:** If  $a_n$  decreasing,  $a_n \geq 0, \lim a_n = 0$ . Then  $\sum (-1)^{n-1} a_n$  convergent.

**Cauchy's Condensation Test:** If  $a_n \geq 0, a_n$  decreasing,  $\Rightarrow [\sum a_n \text{ convergent} \Leftrightarrow \sum 2^n a_{2^n} \text{ also}]$

**Riemann Rearrangement Theorem:** If  $\sum_{k=1}^{\infty} a_k$  is conditionally convergent.  $\Rightarrow \forall s \in \mathbb{R}, \exists$  bijection  $\sigma : \mathbb{N} \rightarrow \mathbb{N} \text{ s.t. } \sum_{n=1}^{\infty} a_{\sigma(n)} = s$ .

**Rearrangement for Abs Series:** If  $\sum_{n=1}^{\infty} a_n$  is absolutely convergent.  $\Rightarrow \forall$  bijection  $\sigma : \mathbb{N} \rightarrow \mathbb{N}, \sum_{n=1}^{\infty} a_{\sigma(n)} = \sum_{n=1}^{\infty} a_n$ .

## 3 Continuous & Differentiable Functions Properties

**Def of lim of func:**  $1. \forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } 0 < |x - x_0| < \delta, \Rightarrow |f(x) - L| < \varepsilon$ .  $2. \forall x_n \neq x_0 \text{ s.t. } \lim(x_n) = x_0, \text{ then } \lim f(x_n) = L$ .

**Def of cont of func:**  $1. \forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } \forall x, |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$ .  $2. \forall (x_n)_{n \in \mathbb{N}}, \text{ if } \lim x_n = x_0 \Rightarrow \lim f(x_n) = f(x_0)$

**Def of differentiable:**  $1. f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$   $2. f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ .

**The Extreme Value Theorem (EVT):** If  $f : [a, b] \rightarrow \mathbb{R}$  is continuous, then  $f$  has a max and min value.

**Intermediate Value Theorem (IVT):**  $f : [a, b] \rightarrow \mathbb{R}$  is continuo,  $f(a) \neq f(b), y \in [f(a), f(b)] \Rightarrow \exists c \in (a, b) \text{ s.t. } f(c) = y$

**IVT (for deriv) / (Darboux Theorem):** If  $f$  differentiable in  $I \Rightarrow \forall m \in (f'(a), f'(b))$  we have:  $\exists c \text{ s.t. } f'(c) = m$

**Rolle's Theorem:** If  $f$  continuous on  $[a, b]$ , differentiable on  $(a, b)$  and  $f(a) = f(b)$  Then:  $\exists c \in (a, b) \text{ s.t. } f'(c) = 0$

**Mean Value Theorem (MVT):** Let  $f, g$  continuous on  $[a, b]$  and differentiable on  $(a, b)$ : ps:  $a=b$  时依然成立

·  $1. \exists c \in (a, b) \text{ s.t. } \frac{f(b) - f(a)}{b - a} = f'(c)$   $2. \exists c \in (a, b) \text{ s.t. } \frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$

**Theorem:** If  $f$  is 1-1 and continuous  $\Rightarrow f$  is strictly monotone. and  $f^{-1}$  is continuous and strictly monotone.

**Inverse Function Theorem:** Let  $f$  differentiable in  $I$  (open interval), and  $\forall x, f'(x) \neq 0$ :

· **1.**  $f(I)$  is open interval. **2.**  $f$  is invertible. **3.**  $f^{-1}$  is differentiable and  $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$

**Second Deriv Test:**  $f$  twice continuously differentiable in  $I, f'(x_0) = 0$ : **1.**  $f''(x_0) > 0 (< 0)$  local min (max) point.

4 Pointwise & Uniformly Convergence

4.1 Pointwise Convergence

Pointwise Convergence: Let  $(f_n)_{n \in \mathbb{N}}$  be a sequence of functions.  $f_n : E \rightarrow \mathbb{R}$  converge pointwise to  $f : E \rightarrow \mathbb{R}$  if: 不加说明默认 pointwise

- 1. Version 1:  $\forall x \in E, \forall \varepsilon > 0, \exists N_{\varepsilon, x}$  s.t.  $\forall n > N \ |f_n(x) - f(x)| < \varepsilon$
- 2. Version 2:  $\forall x \in E, \lim f_n(x)$  exists and  $\lim f_n(x) = f(x)$

Pointwise Convergence NOT Preserve: <sup>1</sup>continuity, <sup>2</sup>differentiability (But converge uniformly), <sup>3</sup> $(\lim f_n)' \neq \lim f'_n$ , <sup>4</sup> $\int \lim f_n \neq \lim \int f_n$

· 对应反例:  $f_n : [0, 1] \rightarrow \mathbb{R} := x^n$   $f_n : (-1, 1) \rightarrow \mathbb{R} := \sqrt{x^2 + \frac{1}{n}}$   $f_n : (-1, 1) \rightarrow \mathbb{R} := \frac{\sin(nx)}{n}$   $f_n : (0, 1] \rightarrow \mathbb{R} := \begin{cases} n, 0 < x \leq \frac{1}{n} \\ 0, \frac{1}{n} < x \leq 1 \end{cases}$

4.2 Uniformly Convergence

Graph:  $\Gamma(f) := \{(x, f(x)) | x \in E\}$  ps:  $f : E \rightarrow \mathbb{R}$

Vertical Neighbourhood  $\mathcal{N}(f, \varepsilon) := \{(x, y) | x \in E, |y - f(x)| < \varepsilon\}$  ps:  $f : E \rightarrow \mathbb{R}$

Uniformly Convergence:  $f_n : E \rightarrow \mathbb{R}$  converge uniformly iff:

1. Version 1:  $\forall \varepsilon > 0, \exists N_\varepsilon$  s.t.  $\forall n > N, \forall x \in E, |f_n(x) - f(x)| < \varepsilon$

2. Version 2:  $\lim [\sup_{x \in E} |f_n(x) - f(x)|] = 0$

3. Version 3 | Uniformly Cauchy:  $\forall \varepsilon > 0, \exists N_\varepsilon \in \mathbb{N}$  s.t.  $\forall x \in E, m, n > N \ |f_n(x) - f_m(x)| < \varepsilon$  || Converge Unif  $\Leftrightarrow$  Unif Cauchy

4. Version 4 | Vertical Neighbourhood Test:  $\forall \varepsilon > 0, \exists N \in \mathbb{N}$  s.t.  $\forall n > N, \Gamma(f_n) \subset \mathcal{N}(f, \varepsilon)$

Sequence Test: If  $f_n : E \rightarrow \mathbb{R}$  converge uniformly to  $f$ .  $\Rightarrow$  If  $(x_n)_{n \in \mathbb{N}} \in E, \lim |f_n(x_n) - f(x_n)| = 0$  不能当作  $\forall x_n$

Properties of Uniformly Convergent: Let  $f_n, f, g : I \rightarrow \mathbb{R}$  Then: NOT preserve: differentiable.

1. Continuous: If  $\lim f_n(x) = f(x)$  uniformly,  $\forall f_n$  continuous at  $x_0 \Rightarrow f$  continuous at  $x_0$ .

$\Downarrow \forall x \in I, (\lim f_n)'(x) = \lim f'_n(x)$

2. Termwise differentiation: If  $I$  is bounded open;  $\forall f_n$  diff,  $\lim f'_n = g$  unif;  $\exists x_0 \in I$  s.t.  $f_n(x_0)$  converges.  $\Rightarrow \lim f_n = f$  unif;  $f'$  diff and  $f' = g$ .  
For Series: If  $I$  is bounded open;  $\forall f_n$  diff,  $\sum f'_n$  exists and unif;  $\exists x_0 \in I$  s.t.  $\sum f_n(x_0)$  converge.  $\Rightarrow \sum f_n$  unif and  $(\sum f)' = \sum f'$

3. Uniform Convergence Theorem: If  $\lim f_n(x) = f(x)$  uniformly on  $E \in \mathcal{M}, m(E) < \infty, f_n \in \mathcal{L}^1(\mathbb{R}) \Rightarrow f \in \mathcal{L}^1(E)$  and  $\lim \int_E f_n(x) dx = \int_E f dx$   
For Series: If  $\sum f_n$  exists and uniformly on  $E \in \mathcal{M}, m(E) < \infty, f_n \in \mathcal{L}^1(\mathbb{R}) \Rightarrow \sum \int f_n \in \mathcal{L}^1(E)$  and  $\int \sum f_n = \sum \int f_n$

4.  $\oplus$  If  $f_n : [a, b] \rightarrow \mathbb{R}, g_n : [a, b] \rightarrow \mathbb{R}$  converge uniformly.  $\Rightarrow \frac{f_n}{g_n}$  converge uniformly also. ps:  $(a, b)$  不行 Hint:  $f_n g - f g_n = f_n g - f g + f g - f g_n$

Def of Series of Functions: Let  $f_n : E \rightarrow \mathbb{R}$ ; Def partial sum  $s_n(x) := \sum_{k=1}^n f_k(x)$

1.  $\sum f_n$  converge pointwise iff  $s_n$  converge pointwise. 2.  $\sum f_n$  converge uniformly iff  $s_n$  converge uniformly. 3.  $\sum f_n$  converge absolutely (pointwise) iff  $\sum |f_n|$  converge (pointwise).

Weierstrass M-Test:  $f_n : E \rightarrow \mathbb{R}, \exists (M_n)_{n \in \mathbb{N}} \geq 0$  s.t.  $\forall x \in E, \forall n \in \mathbb{N} : |f_n| < M_n; \sum M_n$  converges.  $\Rightarrow \sum f_n$  converge absolutely & Uniformly.

5 Power Series

5.1 Power Series, Radius of Convergence, Continuity and Differentiability

Power Series: A series from  $\sum_{n=0}^\infty a_n(x - c)^n$  Center:  $c$ . Coefficients:  $a_n$ .

Radius of Convergence (R): For a power series  $\sum a_n(x - c)^n$ , there are three situations for it's radius of convergence:

1.  $R = 0$  if  $\sum a_n(x - c)^n$  converges only  $x = c$ .

2.  $R = \infty$  if  $\sum a_n(x - c)^n$  converges absolutely  $\forall x \in \mathbb{R}$ .

3.  $\exists R > 0$  s.t.  $\sum a_n(x - c)^n$  converges absolutely if  $|x - c| < R$ ; diverges if  $|x - c| > R$ .

4. All-In-One:  $R = \sup\{r \geq 0 : (a_n r^n)_{n \in \mathbb{N}} \text{ is bounded}\} \in [0, \infty]$

Properties | In Radius of Convergence: Assume  $f(x) = \sum a_n(x - c)^n$  with  $R > 0$ .

1. Continuous:  $f(x)$  continuous on  $(c - R, c + R)$ .

2. Convergence:  $\forall 0 < r < R, f(x)$  converges absolutely & uniformly on  $[c - r, c + r]$ .

3. Preserve R:  $\forall (a_n)_{n \in \mathbb{N}}$ , we have:  $\sum_{n=0}^\infty a_n(x - c)^n$  and  $\sum_{n=1}^\infty n a_n(x - c)^{n-1}$  have same  $R$ . ps: 提供了一种由  $n - 1 \leftrightarrow n$  的方法

4. Termwise differentiation:  $f(x)$  is differentiable on  $(c - R, c + R)$  ||  $f'(x) = \sum_{n=1}^\infty n a_n(x - c)^{n-1}, x \in (c - R, c + R)$   
 $\forall 0 < r < R, f'(x)$  converges absolutely and uniformly on  $[c - r, c + r]$ . ps: 即可以对里面求导

5. Coefficients & Diff:  $f(x)$  is infinitely differentiable on  $(c - R, c + R)$  ||  $a_k = \frac{f^{(k)}(c)}{k!}$

6. Uniqueness: If  $\sum a_n(x - c)^n = \sum b_n(x - c)^n$  for  $\forall x \in \mathbb{R}, |x - c| < r_0 \Rightarrow a_n = b_n$

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5.2 Exponential Function & Real Analytic Function

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Exponential Function:  $E(x) := \sum_{n=0}^\infty \frac{x^n}{n!}$  with  $R = \infty$  ps:  $E(x) = \lim_{n \rightarrow \infty} (1 + \frac{x}{n})^n$  Hint: 证 pointwise 考虑  $\ln$ , as  $\ln(1 + \frac{x}{n}) = \int_1^{1+\frac{x}{n}} \frac{1}{t} dt$ , 缩放, squeeze theorem, uniformly 考虑 unif cont

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Real Analytic Function:  $f : I \rightarrow \mathbb{R}$  is real analytic iff  $\exists (a_n)_{n \in \mathbb{N}}$  s.t.  $f(x) = \sum_{n=0}^\infty a_n(x - c)^n$  for  $\forall x \in I$  i.e. Series converges pointwise on  $I$ .

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Taylor Series: Given <sup>1</sup>Open Interval  $I$ , <sup>2</sup>infinitely differentiable  $f : I \rightarrow \mathbb{R} \Rightarrow$  Taylor Series of  $f$  at  $c \in I := \sum_{n=0}^\infty \frac{f^{(n)}(c)}{n!} (x - c)^n$

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· Not all infinitely differentiable functions are real analytic: e.g.  $f(x) = \{e^{-1/x^2}, x > 0; 0, x < 0$

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Taylor's Theorem: If  $f : (a, b) \rightarrow \mathbb{R}$  is differentiable function at  $x_0 \in (a, b)$ :

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- 1.  $P_{n, f, x_0}(x) := f(x_0) + \sum_{k=1}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$  (Taylor's polynomial of degree  $n$  at  $x_0$ )
- 2. If  $f^{(n+1)}$  exists on  $(a, b)$ , then:  $\exists \xi \in (x, x_0)$  s.t.  $f(x) = P_{n, f, x_0}(x) + \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1}$

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3. (Lagrange) Remainder  $R_{n,f,x_0} := f(x) - P_{n,f,x_0}$ .  $\parallel f : I \rightarrow \mathbb{R}$  is real analytic iff  $\lim R_{n,f,x_0} = 0$  for  $\forall x \in I$

## 6 Open & Closed Sets

**Open Set:** Set  $E$  is open iff  $\forall x \in E, \exists r > 0$ , s.t.  $(x - r, x + r) \subseteq E$

1. **Union of open sets:**  $(E_n)$  are open sets (Infinite).  $\Rightarrow \cup E_n$  is open.
2. **Intersection of open sets:**  $(E_n)$  are open sets (Finite).  $\Rightarrow \cap E_n$  is open.
3. **Find Open Set:**  
If  $f : \mathbb{R} \rightarrow \mathbb{R}$  continuous  $\Rightarrow \{x \in \mathbb{R} : f(x) > \lambda \text{ (or } < \lambda)\}$  is open.
4. **Structure of open sets:**  $\Downarrow^\ominus$  can be disjoint Hint:  $I'_k = \cup_{l=1}^k I'_l$   
 $E$  is a nonempty open set  $\Leftrightarrow E = \cup I_n, I_n$  is open interval. (Unique)

**Closed Set:** Set  $F \subseteq \mathbb{R}$  is closed iff  $\mathbb{R} \setminus F$  is open.

1. **Union of closed sets:**  $(F_n)$  are closed sets (Finite).  $\Rightarrow \cup F_n$  is closed.
2. **Intersection of closed sets:**  $(F_n)$  are closed sets (Infinite).  $\Rightarrow \cap F_n$  is closed.
3. **Find Closed Set:**  
If  $f : \mathbb{R} \rightarrow \mathbb{R}$  continuous  $\Rightarrow \{x \in \mathbb{R} : f(x) \leq \lambda \text{ (or } \geq \lambda) \text{ (or } = \lambda)\}$  is closed.
4. **Sequentially Closed:**  
If  $F$  is closed set,  $\Leftrightarrow \forall (x_n)_{n \in \mathbb{N}} \in F$  convergent.  $x = \lim x_n \in F$

**Open Set  $\rightarrow \cup$  Closed Set:** Nonempty open set  $E \rightarrow$  union of *disjoint closed* intervals. Hint: <sup>1</sup>Structure of open set. <sup>2</sup> each open interval  $(a, b) = \cup[a + \frac{1}{n}, b - \frac{1}{n}]$ . <sup>3</sup> closed set  $\rightarrow$  disjoint.

**Closed Set  $\rightarrow \cap$  Open Set:** Closed set  $F \rightarrow$  intersection of *open* sets. e.g.  $[0, 1] = \cap(0 - \frac{1}{n}, 1 + \frac{1}{n})$

**Unbounded  $\rightarrow \cup$  Bounded:** Any closed / open set  $F \rightarrow$  countable union of *bounded* closed / open sets. (Hint:  $K_n = F \cap [-n, n]$   $K_n = F \cap (-n, n)$ )

**Both Open & Closed:** Only  $\emptyset$  and  $\mathbb{R}$  are both open and closed.

**Some Properties of Sets:**

1. **De Morgan's Laws:**  $(A \cap B)^C = A^C \cup B^C$   $(A \cup B)^C = A^C \cap B^C$   $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$   $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
2.  ${}^\oplus E \setminus F = F^C \cap E$  is open (i.e. open - closed = open)  $\parallel F \setminus E = E^C \cap F$  is open (i.e. closed - open = closed)

## 7 Lebesgue Outer Measure || Lebesgue Measurable || Lebesgue Measure

ps: **Countably Additivity:**  $(E_n)_{n \in \mathbb{N}}$  disjoint sets,  $m$  be measure.  $\Rightarrow$  Countably Additive if:  $m(\cup_n E_n) = \sum m(E_n)$

### 7.1 Lebesgue Outer Measure

**Lebesgue Outer Measure ( $m^*$ ):** Can interpret  $m^*$  as  $m^* : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty)$  ps: 本质是寻找到最小的区间可以覆盖  $E$ , 用前面  $\lambda$  的定义来定义“长度”

1. **Version 1|Open Intervals:**  $m^*(E) := \inf\{\sum_{n=1}^\infty \lambda(I_n) \mid E \subseteq \cup_{n=1}^\infty I_n, I_n \text{ is a open interval}\}$

ps:  $\forall \varepsilon > 0, \exists I_n$  open interval s.t.  $E \subseteq \cup I_n$  and  $m^*(E) < \sum \lambda(I_n) < m^*(E) + \varepsilon$

2. **Version 2|Open Sets:**  $m^*(E) = \inf\{\lambda(U), E \subseteq U, U \text{ is open set}\}$  ps:  $\lambda(U) = \sum \lambda(I_n)$

ps:  $\forall \varepsilon > 0, \exists U$  open set s.t.  $E \subseteq U$  and  $m^*(E) < m^*(U) < m^*(E) + \varepsilon$  or  $m^*(E) < \lambda(U) < m^*(E) + \varepsilon$

· **Remark:** If we cannot find  $(I_n)_{n \in \mathbb{N}}$  s.t.  $\sum \lambda(I_n) < \infty \Rightarrow m^*(E) = \infty$ .

· **Basic 3 Properties:** Let Set  $E, F$  s.t.  $E \subseteq F \subset \mathbb{R}$ .

1. **Null Empty Set:**  $m^*(\emptyset) = 0$ .  $\parallel$  Null Set  $E$ :  $m^*(E) = 0$ .
2. **Monotonicity:**  $m^*(E) \leq m^*(F)$ .
3. **Countable Subadditivity:**  $m^*(\cup_{n=1}^\infty E_n) \leq \sum_{n=1}^\infty m^*(E_n)$ .

**Useful Properties|By Def of inf:** Let  $E$  be a set,  $m^*(E) < \infty$

Then  $\forall \varepsilon > 0, \forall \delta > 0$ :  $\exists (I_n)_{n \in \mathbb{N}}, I_n$  open intervals s.t.:  ${}^\oplus$

<sup>1</sup>  $E \subseteq \cup I_n$  | <sup>2</sup>  $m^*(E) \leq \sum \lambda(I_n) < m^*(E) + \varepsilon$  | <sup>3</sup>  $I_n \cap E \neq \emptyset, \forall n$  | <sup>4</sup>  $\lambda(I_n) < \delta, \forall n$

<sup>5</sup>  $I_n \cap E \not\subseteq I_k, \forall n \neq k$ .  $\leftarrow$  无覆盖项 无无关项  $\uparrow$  ps: 也使用与是 open set 的.

· **More Properties:**

1. **Interval:** If  $J \subseteq \mathbb{R}$  is interval.  $m^*(J) = \lambda(J)$ .
2. **Countable:** Any Countable set  $E \Rightarrow m^*(E) = 0$   $\exists m^*(E) = 0, E$  uncountable (e.g. Cantor Set (Closed Set))
3. **Further| $m^{**}(E)$ :**  $m^*(E) = m^{**}(E) := \inf\{\sum \lambda(I_n) \mid E \subseteq \cup I_n, I_n \text{ interval}\}$   $\parallel$   ${}^\oplus m^*(E) \leq \sum \lambda(I_n)$ , if  $E = \cup I_n, I_n$  intervals.
4. **Translation Invariance:**  $m^*(E) = m^*(x + E)$

**Def of dist(E, F):**  $\text{dist}(E, F) := \inf\{|x - y| : x \in E, y \in F\}$

**Lemma:** <sup>1</sup>  $E, F$  are nonempty closed sets; <sup>2</sup>  $E \cap F = \emptyset$ ; <sup>3</sup>  $E$  bounded.  $\Rightarrow \text{dist}(E, F) > 0$

**Countable Additivity|Outer Measure:**

1. **Basic:** If  $\text{dist}(E, F) > 0 \Rightarrow m^*(E \cup F) = m^*(E) + m^*(F)$ .
2. **Closed|Open|Interval Sets:** If  $(K_n)_{n \in \mathbb{N}}$  is disjoint, (Closed (bdd) | Open | Interval), sets  $\Rightarrow m^*(\cup_n K_n) = \sum_{n=1}^\infty m^*(K_n)$

### 7.2 Lebesgue Measurable

**Lebesgue Measurable:**

1. **Version 1:** A set  $E \subseteq \mathbb{R}$  is *Lebesgue measurable* if  $\forall \varepsilon > 0, \exists$  open set  $U \subseteq \mathbb{R}$  s.t.  $E \subseteq U$  and  $m^*(U \setminus E) < \varepsilon$  ps: 被 open set 从外“恰好”包住
2. **Version 2:** A set  $E \subseteq \mathbb{R}$  is *Lebesgue measurable* if  $\forall \varepsilon > 0, \exists$  closed set  $U \subseteq \mathbb{R}$  s.t.  $U \subseteq E$  and  $m^*(E \setminus U) < \varepsilon$  ps: 被 closed set 从内“恰好”包住

·  $\mathcal{M}$  := collection of all Lebesgue measurable sets of  $\mathbb{R}$ .

**Property of Lebesgue Measurable Sets:**

1. **Star-Closed Set|Open Set|Interval:** Any closed set / open set / interval is Lebesgue measurable.
2. **Union  $\cup$ |Intersection  $\cap$ :** If  $(E_n)_{n \in \mathbb{N}}$  is Lebesgue Measurable  $\Rightarrow$  Also: <sup>1</sup>  $E := \cup_{n=1}^\infty E_n$  <sup>2</sup>  $E := \cap_{n=1}^\infty E_n$ . Hint:  $(\cap E_n)^C = \cup E_n^C$
3. **Complement  $E^C$ :** if  $E \subseteq \mathbb{R}$  is Lebesgue Measurable  $\Rightarrow \mathbb{R} \setminus E$  is also Lebesgue Measurable.
4. **Measure Zero Set:**  $m^*(E) = 0 \Leftrightarrow E$  Lebesgue measurable.
5. **Translation Invariance:**  $m(E) = m(x + E)$   $\parallel$  If  $E$  is Lebesgue Measurable  $\Rightarrow x + E$  also.
6. **Find Lebesgue Measurable Set:** A simple function  $s \Rightarrow \{x \in \mathbb{R}, s(x) > \lambda (< \lambda \text{ or } = \lambda)\}$  is Lebesgue Measurable.

### 7.3 Lebesgue Measure

**Lebesgue Measure:** *Lebesgue Measure* is a function  $m : \mathcal{M} \rightarrow [0, \infty)$  s.t.  $m(E) := m^*(E), \forall m \in \mathcal{M}$

ps: 原来是  $m^* : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty)$ , 现在是  $m : \mathcal{M} \rightarrow [0, \infty)$

• **Property:** ★ All properties of Lebesgue Outer Measure hold for Lebesgue Measure. ★

★ **Countable Additivity for Measure:** If  $(E_n)_{n \in \mathbb{N}}$  disjoint & Lebesgue Measurable.  $\Rightarrow m(\cup_{n=1}^{\infty} E_n) = \sum_{n=1}^{\infty} m(E_n)$

**Monotone Convergence for Measure:** If  $(E_n)_{n \in \mathbb{N}}$  Lebesgue Measurable.

1. **Increasing  $\uparrow$ :** If  $E_n \subseteq E_{n+1} \Rightarrow m(\cup_{n=1}^{\infty} E_n) = \lim_{n \rightarrow \infty} m(E_n)$
2.  **$\oplus$  decreasing  $\downarrow$ :** If  $E_{n+1} \subseteq E_n, m(E_1) < \infty \Rightarrow m(\cap_{n=1}^{\infty} E_n) = \lim_{n \rightarrow \infty} m(E_n)$

(Hint:  $F_n = E_1 \setminus E_n$  (1)  $m(F_n) = m(E_1) - m(E_n)$   $m(\cup F_n) = \lim m(F_n) = m(E_1) - m(\cap E_n)$ ,  $m(\cup F_n) = m(E_1 \setminus \cap E_n) = m(\cap E_n) + m(\cup F_n)$ )

**Vitali Set:** <sup>1</sup> 存在 not Lebesgue measurable set. <sup>2</sup> 存在 not countability additive set.

• **Example:**  $\forall$  cosets  $B \in \{x + \mathbb{Q} | x \in \mathbb{R}\}$ , choose a number  $x_B \in B \cap [0, 1]$   $A := \{x_B : x_B \in B\}$  <sup>1</sup> A is not Lebesgue Measurable Set. <sup>2</sup>  $E := \cup_{q \in \mathbb{Q} \cap [-1, 1]} (q + A)$  not Countability Additive.

## 8 Lebesgue Integral

### 8.1 Step Function || Characteristic Function || Definition of Integral

**Characteristic Function:**  $\chi_E(x) := \{1, x \in E ; 0, x \notin E ; E \subset \mathbb{R}$

**Basic Integral Definitions:**  $\int_{\mathbb{R}} \chi_I(x) dx = m(I) = \lambda(I)$  ||  $\int_{\mathbb{R}} c \chi_I(x) dx = c \lambda(I)$  ||  $\int_{\mathbb{R}} \sum_{j=1}^m c_j \chi_{I_j}(x) dx = \sum_{j=1}^m c_j \lambda(I_j), c_j \geq 0$

**Step Function:**  $f : \mathbb{R} \rightarrow [0, \infty)$  is a (non-negative) step function if  $f(x) = \sum c_j \chi_{I_j}(x)$  for some  $c_j$  and  $\chi_{I_j}$

• **Property:** If  $f$  is a step function  $\Rightarrow \exists c_n, I_n$  s.t.  $I_n$  disjoint and  $f(x) = \sum c_n \chi_{I_n}(x)$

**Defined of  $\int f dx$ :**  $\int_{\mathbb{R}} f$  is defined iff: It's Lebesgue integrable or  $f : \mathbb{R} \rightarrow (\pm)[0, \infty)$  is Lebesgue Measurable. 本质是不允许  $(\infty - \infty)$

### 8.2 Simple Functions || Measurable Functions

#### 8.2.1 Simple Functions & Measurable Function

**Simple Function:**  $s : \mathbb{R} \rightarrow \mathbb{R}$  is a simple function iff:  $s(x) = \sum_{j=1}^k c_j \chi_{E_j}, E_j$  are Lebesgue Measurable sets. ||  $E_j$  can be disjoint.

**Integral of Simple Function|Positive:**  $\int_{\mathbb{R}} s(x) dx := \sum_{j=1}^k c_j m(E_j)$  ||  $\int_{\mathbb{R}} \alpha s_1 + \beta s_2 = \alpha \int_{\mathbb{R}} s_1 + \beta \int_{\mathbb{R}} s_2$

**Measurable Function:**  $f : \mathbb{R} \rightarrow \mathbb{R}$  is Lebesgue Measurable if  $\forall \lambda \in \mathbb{R}, \{x \in \mathbb{R} : f(x) > \lambda\}$  (**Superlevel Set**) is Lebesgue Measurable.

1. **MF|Simple|Continuous Function:** (Simple|Continuous) Function is Lebesgue Measurable. ↓ Hint:  $\{x | f(x) < \lambda\}^C = \{x | f(x) > \lambda - \frac{1}{n}\}$
2.  **$\ominus$  MF|Monotone:** All Monotone functions are Lebesgue Measurable. Hint: Prove it by using definition directly.
3. **MF|Measurable Set:** If  $f$  is Lebesgue Measurable,  $\Rightarrow \{x \in \mathbb{R} : f(x) > \lambda (< \lambda \text{ or } = \lambda)\}$  is Lebesgue Measurable.
4. **MF|property:** If  $f, g, f_n$  is Lebesgue Measurable,  $\Rightarrow \max\{f, g\}, \min\{f, g\}, f \cdot \chi_E, \alpha f \pm \beta g, f^2, f \cdot g, \sup f_n, \inf f_n, \limsup f_n, \liminf f_n$  also.
5. **Measurable Func  $\Rightarrow \exists$  Simple Func:**  $f$  is 非负勒贝格可测函数  $\Rightarrow \exists$  simple functions  $(s_n)_{n \in \mathbb{N}}, 0 \leq s_n \leq s_{n+1} (\uparrow)$  s.t.  $s_n \rightarrow f$  pointwise.

#### 8.2.2 Measurable Functions|Positive

**Integral of Measurable Function|Positive:** Let  $f : \mathbb{R} \rightarrow [0, \infty)$  be Lebesgue Measurable Function.

1. **Basic Def:**  $\int_{\mathbb{R}} f(x) dx := \sup\{\int_{\mathbb{R}} s(x) dx : s \text{ simple}, 0 \leq s \leq f\}$
2. **Extend Def:** If  $f : \mathbb{R} \rightarrow \mathbb{R}$  Lebesgue Measurable,  $E$  Measurable Set.  $\Rightarrow \int_E f(x) dx := \int_{\mathbb{R}} f(x) \chi_E(x) dx$

**Basic Properties of Integral|Positive:** Let  $f, g : [0, \infty) \rightarrow \mathbb{R}$  be non-negative Lebesgue Measurable functions.

1. **★Monotonicity:** If  $0 \leq f(x) \leq g(x) \Rightarrow 0 \leq \int_{\mathbb{R}} f(x) dx \leq \int_{\mathbb{R}} g(x) dx$
2. **Markov's Inequality:** For all  $\lambda > 0 \Rightarrow m(\{x \in \mathbb{R} : f(x) > \lambda\}) \leq \frac{1}{\lambda} \int_{\mathbb{R}} f(x) dx$  函数积分面积 > 其对应底下矩形面积
3.  **$\oplus$  Measure Zero:** If  $E$  s.t.  $m(E) = 0 \Rightarrow \int_E f(x) dx = 0$  Hint:  $\forall$  simple  $s, 0 \leq s \leq f \Rightarrow \int_E s \chi_E = \int_{\mathbb{R}} c_n \chi_{E_n \cap E} = \sum c_n m(E_n \cap E) \leq \sum c_n m(E) = 0$

**MCT for Integral|Positive:** <sup>1</sup>  $f_n : [0, \infty) \rightarrow \mathbb{R}$  non-decreasing (↑) MF; <sup>2</sup>  $\lim f_n = f$  pointwise.  $\Rightarrow$  <sup>1</sup>  $f$  is MF; <sup>2</sup>  $\int_{\mathbb{R}} \lim f_n dx = \lim \int_{\mathbb{R}} f_n(x) dx$

**For Series:** <sup>1</sup>  $f_n : [0, \infty) \rightarrow \mathbb{R}$  MF; <sup>2</sup>  $\sum f_n = f$  pointwise  $\Rightarrow$  <sup>1</sup>  $f$  is MF; <sup>2</sup>  $\int_{\mathbb{R}} \sum f_n = \sum \int_{\mathbb{R}} f_n$

**More Properties of Integral|Positive:** Let  $f, g : [0, \infty) \rightarrow \mathbb{R}$  be non-negative Lebesgue Measurable functions.

1. **Linear Combination:**  $\alpha, \beta > 0 \Rightarrow \int_{\mathbb{R}} \alpha f + \beta g = \alpha \int_{\mathbb{R}} f + \beta \int_{\mathbb{R}} g$
2. **Union of Disjoint:**  $E, F$  disjoint measurable sets  $\Rightarrow \int_{E \cup F} f = \int_E f + \int_F f$
3. **(Reverse) Fatou Lemma:**  $f_n$  non-negative MF  $\Rightarrow \int_{\mathbb{R}} \liminf f_n \leq \liminf \int_{\mathbb{R}} f_n \leq (\limsup \int_{\mathbb{R}} f_n \leq \int_{\mathbb{R}} \limsup f_n)$

### 8.3 Integrating Real-Valued Functions

**Isolated Functions:** For a function  $f$ . We define that  $f^+ := \max\{f, 0\}, f^- := -\min\{f, 0\} \Rightarrow f = f^+ - f^-, |f| = f^+ + f^-$ .

• **Remark:** If  $a < 0: (af)^+ = (-a)f^-$  and  $(af)^- = -af^+$

**Lebesgue Integrable:** A MF  $f : \mathbb{R} \rightarrow \mathbb{R}$  is Lebesgue Integrable iff  $\int_{\mathbb{R}} |f(x)| dx < \infty \Leftrightarrow \int_{\mathbb{R}} f^+ dx < \infty \& \int_{\mathbb{R}} f^- dx < \infty$

1. **Def of Integral:** If  $f \in \mathcal{L}^1(\mathbb{R}) \Rightarrow \int_{\mathbb{R}} f(x) dx := \int_{\mathbb{R}} f^+(x) dx - \int_{\mathbb{R}} f^-(x) dx$
2. **Extended Def:** If  $f : E \rightarrow \mathbb{R}$  with  $E$  Measurable Set s.t.  $\tilde{f} := f(x) \chi_E$  is Lebesgue Integrable  $\Rightarrow \int_E f(x) dx := \int_{\mathbb{R}} \tilde{f}(x) dx$
3.  $\mathcal{L}^1(E) :=$  Collection of all Lebesgue Integrable functions on  $E$ .

**Common LI Functions:**  $\int_1^{\infty} \frac{1}{x^p} dx < \infty$  iff  $p > 1$   $\int_{\mathbb{R}} \frac{c_1}{c_2 + c_3 |x|^p} dx < \infty$  if  $p > 1; c_2, c_3 > 0$   $|\frac{c_1}{c_2 + c_3 |x|^p}| \leq \frac{|c_1|}{|c_2|} \chi_{[-1, 1]} + \frac{|c_1|}{|c_3| |x|^p} \chi_{(-\infty, -1) \cup (1, \infty)}$

If  $p \geq 0, \int_{\mathbb{R}} \frac{1}{(1+|x|)^p} dx < \infty$  iff  $p > 1$  If  $p \geq 0, \int_{\mathbb{R}} \frac{c_1}{(c_2 + c_3 |x|)^p} dx < \infty$  iff  $p > 1, c_2, c_3 > 0$   $\frac{c_1}{(c_2 + c_3 |x|)^p} = \frac{c_1}{c_2^p} \frac{1}{(1 + \frac{c_3}{c_2} |x|)^p}$

If  $p \geq 0, \int_{\mathbb{R}} \frac{1}{x^p} \chi_{(0, 1]} < \infty$  iff  $0 \leq p < 1$

**Property of Integral:** Let  $f, g \in \mathcal{L}^1(\mathbb{R})$ . Then:

- Triangular Inequality for Integrals:**  $|\int_{\mathbb{R}} f(x)dx| \leq \int_{\mathbb{R}} |f(x)|dx$
- Linearity of the Integral:**  $\alpha f + \beta g \in \mathcal{L}^1(\mathbb{R})$  and  $\int_{\mathbb{R}} \alpha f + \beta g = \alpha \int_{\mathbb{R}} f + \beta \int_{\mathbb{R}} g \Rightarrow \mathcal{L}^1(\mathbb{R})$  is a vector space
- Union of Disjoint:**  $E, F$  disjoint measurable sets  $\Rightarrow \int_{E \cup F} f = \int_E f + \int_F f$  (By Linearity of Integral)
- 'Comparison Test':** If  $f: \mathbb{R} \rightarrow \mathbb{R}$  measurable  $\int_{\mathbb{R}} g(x) < \infty$   $|f| < g \Rightarrow f \in \mathcal{L}^1(\mathbb{R})$
- Remark:** Can use p-test, Comparison test ... when we want to prove a function is Lebesgue Integrable.

**Dominated Convergence Theorem (DCT):**  $f_n$  is MF.  $f_n \rightarrow f$  pointwise.  $\exists g \in \mathcal{L}^1(\mathbb{R})$  s.t.  $|f_n| \leq g \Rightarrow f \in \mathcal{L}^1(\mathbb{R})$  &  $\lim \int_{\mathbb{R}} f_n = \int_{\mathbb{R}} f$

**Uniform Convergence Theorem:** If  $\lim f_n(x) = f(x)$  uniformly on  $E \in \mathcal{M}, m(E) < \infty, f_n \in \mathcal{L}^1(\mathbb{R}) \Rightarrow f \in \mathcal{L}^1(E)$   $\lim \int_E f_n(x)dx = \int_E f dx$

**Fundamental Theorem of Calculus (FTC):**  $f: [a, b] \rightarrow \mathbb{R}$  be continuous.  $\Rightarrow F(x) := \int_a^x f(y)dy$  is differentiable  $F'(x) = f(x)$

**Corollary:**  $f: [a, b] \rightarrow \mathbb{R}$  be continuously differentiable.  $\Rightarrow \int_a^b f'(x)dx = f(b) - f(a)$

**Remark:**  $f$  is Lebesgue Integrable. (By Extreme value theorem) and. 需要闭区间 (Closed) 否则需要结合 MCT 或 DCT 极限逼近.

- Integration-By-Parts:** If  $f, g: [a, b] \rightarrow \mathbb{R}$  be continuously differentiable.  $\Rightarrow \int_a^b f g' dx = [fg]_a^b - \int_a^b f' g dx$
- Integration-By-Substitution:** If  $f: [c, d] \rightarrow \mathbb{R}$  continuous &  $\phi: [a, b] \rightarrow [c, d]$  continuous diff  $\Rightarrow \int_a^b f(\phi)\phi' dx = \int_{\phi(a)}^{\phi(b)} f(y)dy$

9 Appendix  
Inequation:

|                               |                                                                                                   |                                                                             |                                                                                                                           |
|-------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Triangular</b>             | $ x + y  \leq  x  +  y $                                                                          | $ x - y  \geq  x  -  y $                                                    | $  x  -  y   \leq  x \pm y .$                                                                                             |
| <b>AM-GM &amp; Cauchy</b>     | $a + b \geq 2\sqrt{ab}, a > 0, b > 0$                                                             | $a^2 + b^2 \geq 2ab$                                                        | $(\sum a_n b_n)^2 \leq \sum a_n^2 \sum b_n^2$                                                                             |
| <b>Bernoulli's Inequality</b> | $(1 + \delta)^\alpha \leq 1 + \delta\alpha, \alpha \in (0, 1], \delta \geq -1$                    |                                                                             | $(1 + \delta)^\alpha \geq 1 + \delta\alpha, \alpha \in [1, +\infty), \delta \geq -1$                                      |
| $e^x, \ln(x)$                 | $e^x \geq 1 + x \geq x$ Hint: Def/MVT                                                             | $\ln x \leq x - 1 \leq x$ Hint: MVT                                         | $e^x \geq 1 + x + \frac{x^2}{2} + \dots$ Hint: Def $\frac{1}{k} \leq \ln(x) - \ln(x-1)$ Hint: $\int_{k-1}^k \frac{dx}{x}$ |
| 常见变换技巧                        | $(1 + \frac{x}{n})^n \geq 1 + n \cdot \frac{x}{n} + C_n^2 (\frac{x}{n})^2$ Hint: Binomial Theorem |                                                                             | $ a_n  x - c ^n =  a_n \rho^n \cdot \frac{ x-c ^n}{\rho^n},  x - c  < \rho < R$                                           |
| $\sin(x), \cos(x)$            | $ \sin(x)  \leq  x $ Hint: MVT                                                                    | $1 - \cos(x) <  x $ Hint: MVT                                               |                                                                                                                           |
| 常见变换技巧 2                      | If $\forall \varepsilon > 0$ $a \leq b + \varepsilon \Rightarrow a \leq b$                        | $\forall \delta, 0 < \delta < 1$ and $\delta a \leq b \Rightarrow a \leq b$ |                                                                                                                           |

$\oplus$  和差化积/积化和差:  $s+s=sc, s-s=cs, c+c=cc, c-c=-ss$  || e.g.  $\sin A + \sin B = 2\sin(\frac{A+B}{2})\cos(\frac{A-B}{2})$   $\sin A \sin B = -\frac{1}{2}[\cos(A+B) - \cos(A-B)]$  Hint: 和差角公式相加减

$\sin(2x) = 2\sin(x)\cos(x)$   $\cos(2x) = 2\cos^2(x) - 1 = 1 - \sin^2(x) = \cos^2(x) - \sin^2(x)$   $\cos(x) = \sqrt{\frac{1+\cos(2x)}{2}}$   $\sin(x) = \sqrt{\frac{1-\cos(2x)}{2}}$

**Hyperbolic Func:**  $\sinh(x) = \frac{e^x - e^{-x}}{2}$   $\cosh(x) = \frac{e^x + e^{-x}}{2}$   $\cosh^2(x) - \sinh^2(x) = 1$   $[\sinh(x)]' = \cosh(x)$   $[\cosh(x)]' = \sinh(x)$

$\sinh(x \pm y) = \sinh(x)\cosh(y) \pm \cosh(x)\sinh(y)$   $\cosh(x \pm y) = \cosh(x)\cosh(y) \pm \sinh(x)\sinh(y)$

**Common Int:**  $\int \tan(x)dx = -\ln|\cos(x)|$ ;  $\int \cot(x)dx = \ln|\sin(x)|$ ;  $\int \sec(x)dx = \ln|\sec(x) + \tan(x)|$ ;  $\int \csc(x)dx = \ln|\csc(x) - \cot(x)| = \ln|\tan(\frac{x}{2})|$

**Common Der:**  $[\tan(x)]' = \sec^2(x)$ ;  $[\cot(x)]' = -\csc^2(x)$ ;  $[\sec(x)]' = \sec(x)\tan(x)$ ;  $[\csc(x)]' = -\csc(x)\cot(x)$ ;  $[\arctan(x)]' = \frac{1}{1+x^2}$ .

$1 + \tan^2(x) = \sec^2(x)$ ;  $1 - \csc^2(x) = \cot^2(x)$ ;  $\csc^2(x) - \cot^2(x) = 1$ .  $[\arcsin(x)]' = \frac{1}{\sqrt{1-x^2}}$ ;  $[\arccos(x)]' = -\frac{1}{\sqrt{1-x^2}}$

**Useful Counterexample:**  $(1/2)^{\frac{1}{n}} \subseteq [1/2, 1)$  **Countably Approximation Tools:**  $\sum_{n=1}^\infty 2^{-n}\varepsilon = \varepsilon$

$q^n - 1: q^n - 1 = (q - 1)(1 + q + q^2 + \dots + q^{n-1})$  ||  $1 \pm x^3: 1 \pm x^3 = (1 \pm x)(1 \mp x + x^2)$

**For exponential:**  $(1 + \frac{x}{n})^n \geq 1 + n(\frac{x}{n}) + \binom{n}{2}(\frac{x}{n})^2$  by Binomial Theorem.

**Useful Limits:** If  $0 \leq x < 1$ , then  $\lim_{n \rightarrow \infty} n^k x^n = 0$

**Common Power Series:**

| $f(x)$              | Taylor                                              | Series                                                  | $R$            | $f(x)$              | Taylor                                          | Series                                            | $R$      |
|---------------------|-----------------------------------------------------|---------------------------------------------------------|----------------|---------------------|-------------------------------------------------|---------------------------------------------------|----------|
| $\frac{1}{1-x}$     | $\sum_{n=0}^\infty x^n$                             | $1 + x + x^2 + x^3 + \dots$                             | 1              | $\frac{1}{(1-x)^2}$ | $\sum_{n=1}^\infty nx^{n-1}$                    | $1 + 2x + 3x^2 + 4x^3 + \dots$                    | 1        |
| $\frac{2}{(1-x)^3}$ | $\sum_{n=2}^\infty n(n-1)x^{n-2}$                   | $2 + 6x + 12x^2 + 20x^3 + \dots$                        | 1              | $e^x$               | $\sum_{n=0}^\infty \frac{x^n}{n!}$              | $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ | $\infty$ |
| $\ln(1+x)$          | $\sum_{n=1}^\infty (-1)^{n+1} \frac{x^n}{n}$        | $x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$             | 1              | $-\ln(1-x)$         | $\sum_{n=1}^\infty \frac{x^n}{n}$               | $x + \frac{x^2}{2} + \frac{x^3}{3} + \dots$       | 1        |
| $\sin x$            | $\sum_{n=0}^\infty (-1)^n \frac{x^{2n+1}}{(2n+1)!}$ | $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$           | $\infty$       | $\cos x$            | $\sum_{n=0}^\infty (-1)^n \frac{x^{2n}}{(2n)!}$ | $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$     | $\infty$ |
| $\arctan x$         | $\sum_{n=0}^\infty (-1)^n \frac{x^{2n+1}}{2n+1}$    | $x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$             | 1              | $\sinh x$           | $\sum_{n=0}^\infty \frac{x^{2n+1}}{(2n+1)!}$    | $x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$     | $\infty$ |
| $\cosh x$           | $\sum_{n=0}^\infty \frac{x^{2n}}{(2n)!}$            | $1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$           | $\infty$       | $(1+x)^k$           | $\sum_{n=0}^\infty \binom{k}{n} x^n$            | $1 + kx + \frac{k(k-1)x^2}{2!} + \dots$           | 1        |
| $\ln x$             | $\sum_{n=1}^\infty (-1)^{n+1} \frac{(x-1)^n}{n}$    | $(x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \dots$ | $1, 0 < x < 2$ | $\frac{1}{1+x}$     | $\sum_{n=0}^\infty (-1)^n x^n$                  | $1 - x + x^2 - x^3 + \dots$                       | 1        |

**Sum of n, n<sup>2</sup>, n<sup>3</sup>:**  $\sum_{k=1}^n k = \frac{n(n+1)}{2}$   $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$   $\sum_{k=1}^n k^3 = [\frac{n(n+1)}{2}]^2$

**Interchanging the Order of Summation:**  $\sum_{n=1}^p \sum_{m=1}^q f(n, m) = \sum_{m=1}^q \sum_{n=1}^p f(n, m)$  ||  $\oplus \sum_{k=0}^n \sum_{i=0}^k = \sum_{i=0}^n \sum_{k=i}^n$  Hint: 列出来看 or 画图

$\oplus$  **Summation By Parts:**  $\sum_{k=1}^n a_k(b_{k+1} - b_k) = (a_{n+1}b_{n+1} - a_1b_1) - \sum_{k=1}^n (a_{k+1} - a_k)b_{k+1}, \forall (a_k)_{k=1}^{n+1}, (b_k)_{k=1}^{n+1}$

$\Rightarrow \sum_{k=1}^n a_k b_k = a_{n+1}\sigma_{n+1} - \sum_{k=1}^n (a_{k+1} - a_k)\sigma_{k+1}, \forall (a_k)_{k=1}^{n+1}, (b_k)_{k=1}^{n+1}$   $\sigma_k := \sum_{j=1}^{k-1} b_j$

**Lebesgue Integrable:**  $f$  continuous +  $f$  in bounded interval  $\Rightarrow f \in \mathcal{L}^1$

Related to Metric:

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>If <math>a_n \uparrow</math>, then <math>\lim_{n \rightarrow \infty} a_n = \sup a_n</math></div> <div>If <math>a_n \downarrow</math>, then <math>\lim_{n \rightarrow \infty} a_n = \inf a_n</math></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <div><math>\limsup_{n \rightarrow \infty} a_n = \inf_{n \geq 1} \sup_{k \geq n} a_k</math></div> <div><math>\liminf_{n \rightarrow \infty} a_n = \sup_{n \geq 1} \inf_{k \geq n} a_k</math></div>                                                                                                                                                                                           | <div><math>\limsup_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k</math></div> <div><math>\liminf_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k</math></div> <div><math>A_n</math> sets</div> |
| <div><math>\sup_{n \in \mathbb{N}} f_n(x) := \sup\{f_n(x) : n \in \mathbb{N}\}</math></div> <div><math>\{x : \sup_{n \in \mathbb{N}} f_n(x) &gt; \lambda\} = \bigcup_{n \in \mathbb{N}} \{x : f_n(x) &gt; \lambda\}</math></div> <div><math>\{x : \limsup_{n \in \mathbb{N}} f_n(x) &gt; \lambda\} = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} \{x : f_k(x) &gt; \lambda\}</math></div> <div>特殊集合构造技巧 I: <math>\{x \in \mathbb{R} : (f + g)(x) &gt; \lambda\} = \bigcup_{t \in \mathbb{Q}} (\{x \in \mathbb{R} : f(x) &gt; \lambda - t\} \cap \{x \in \mathbb{R} : g(x) &gt; t\})</math></div> <div>特殊集合构造技巧 II: <math>\{x \in \mathbb{R} : \forall \varepsilon &gt; 0, \exists N &gt; 0 \text{ s.t. } \dots &lt; \varepsilon\} = \bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \{x \in \mathbb{R} : \dots &lt; \frac{1}{k}\}</math></div> | <div><math>\inf_{n \in \mathbb{N}} f_n(x) := \inf\{f_n(x) : n \in \mathbb{N}\}</math></div> <div><math>\{x : \inf_{n \in \mathbb{N}} f_n(x) &gt; \lambda\} = \bigcap_{n \in \mathbb{N}} \{x : f_n(x) &gt; \lambda\}</math></div> <div><math>\{x : \liminf_{n \in \mathbb{N}} f_n(x) &gt; \lambda\} = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} \{x : f_k(x) &gt; \lambda\}</math></div> |                                                                                                                                                                                                                                                                |

**Def of function:** for a function  $f$  well-defined on a set  $E$ , we have  $f(x) < \infty$  for all  $x \in E$

**Cauchy-Schwarz Inequality:**  $\forall x, y \in \mathbb{R}^n \Rightarrow (\sum_{i=1}^n x_i y_i)^2 \leq (\sum_{i=1}^n x_i^2)(\sum_{i=1}^n y_i^2)$